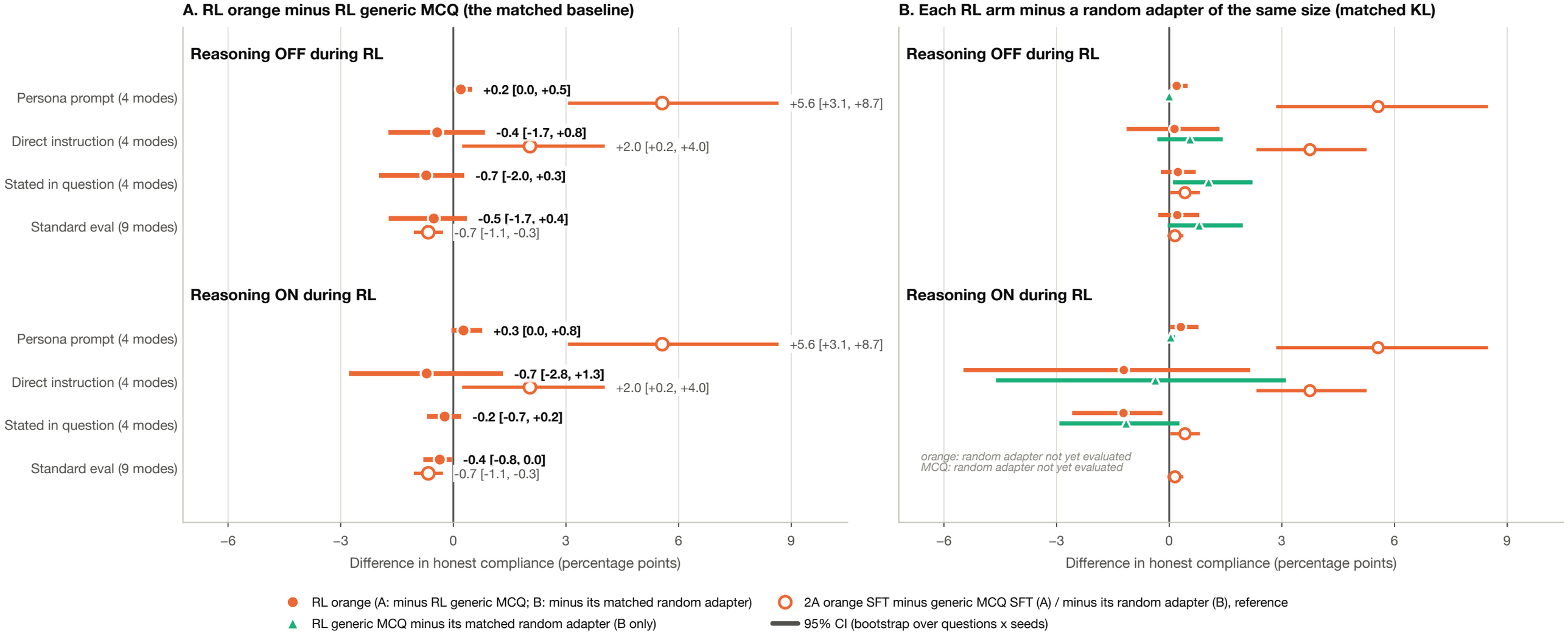
Orange RL does no better than the RL generic-MCQ baseline, or than random adapters of equal size



CAVEAT: the generic-MCQ reward is already about 0.95 at step 0 and saturates, so RL generic MCQ is an under-dosed baseline: prompt groups that carried gradient (of 2,400) were 64 (reasoning off) and 108 (on) for generic MCQ versus 89 and 198 for orange (fig3, fig5). By KL(base || adapter) the two RL arms moved the model by comparable amounts (fig3; KL is measured without the persona prompt), so panel B, equal-size random adapters, is the dose-matched reference.

Qwen3.6-27B, test split (500 questions), final checkpoint (step 50), 5 seeds per RL arm vs 3 random adapters; 95% CI by bootstrap over questions and seeds. Panel B pools only the modes finished for all 3 random adapters (the RL arm is restricted to the same modes); 'not yet evaluated' = no complete block yet. 2A SFT reference: reasoning-off SFT; 2A has no generic-MCQ SFT run on the 4-mode stated block, and its standard block used all 9 modes per question (RL: one random mode per question).